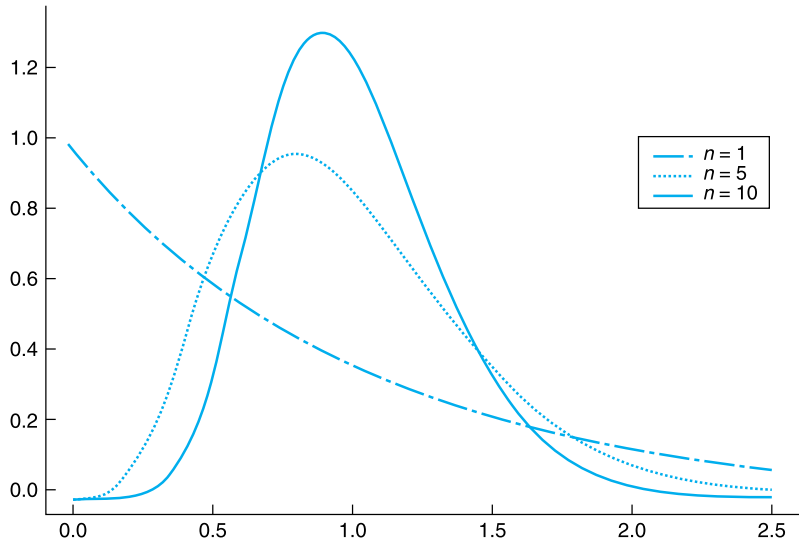


FIGURE 6.3
Densities of the average of n exponential random variables having mean 1.

6.3.2 How large a sample is needed?

The central limit theorem leaves open the question of how large the sample size n needs to be for the normal approximation to be valid, and indeed the answer depends on the population distribution of the sample data. For instance, if the underlying population distribution is normal, then the sample mean $\overline{X}$ will also be normal regardless of the sample size. A general rule of thumb is that one can be confident of the normal approximation whenever the sample size n is at least 30. That is, practically speaking, no matter how nonnormal the underlying population distribution is, the sample mean of a sample of size at least 30 will be approximately normal. In most cases, the normal approximation is valid for much smaller sample sizes. Indeed, a sample of size 5 will often suffice for the approximation to be valid. Figure 6.3 presents the distribution of the sample means from an exponential population distribution for samples of sizes $n = 1, 5, 10$.

6.4 The sample variance

Let $X_1, \ldots, X_n$ be a random sample from a distribution with mean μ and variance σ^2. Let $\overline{X}$ be the sample mean, and recall the following definition from Section 2.3.2.

Definition. The statistic S^2, defined by

$$S^2 = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{n - 1}$$

is called the *sample variance*. $S = \sqrt{S^2}$ is called the *sample standard deviation*.

To compute $E[S^2]$, we use an identity that was proven in Section 2.3.2: For any numbers $x_1, \ldots, x_n$

$$\sum_{i=1}^{n}(x_i - \overline{x})^2 = \sum_{i=1}^{n} x_i^2 - n\overline{x}^2$$

where $\overline{x} = \sum_{i=1}^{n} x_i/n$. It follows from this identity that

$$(n-1)S^2 = \sum_{i=1}^{n} X_i^2 - n\overline{X}^2$$

Taking expectations of both sides of the preceding yields, upon using the fact that for any random variable W, $E[W^2] = \mathrm{Var}(W) + (E[W])^2$,

$$\begin{aligned}(n-1)E[S^2] &= E\left[\sum_{i=1}^{n} X_i^2\right] - nE[\overline{X}^2] \\ &= nE[X_1^2] - nE[\overline{X}^2] \\ &= n\mathrm{Var}(X_1) + n(E[X_1])^2 - n\mathrm{Var}(\overline{X}) - n(E[\overline{X}])^2 \\ &= n\sigma^2 + n\mu^2 - n(\sigma^2/n) - n\mu^2 \\ &= (n-1)\sigma^2\end{aligned}$$

or

$$E[S^2] = \sigma^2$$

That is, the expected value of the sample variance S^2 is equal to the population variance σ^2.

6.5 Sampling distributions from a normal population

Let $X_1, X_2, \ldots, X_n$ be a sample from a normal population having mean μ and variance σ^2. That is, they are independent and $X_i \sim \mathcal{N}(\mu, \sigma^2)$, $i = 1, \ldots, n$. Also let

$$\overline{X} = \sum_{i=1}^{n} X_i/n$$

and

$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}$$

denote the sample mean and sample variance, respectively. We would like to compute their distributions.

6.5.1 Distribution of the sample mean

Since the sum of independent normal random variables is normally distributed, it follows that $\overline{X}$ is normal with mean

$$E[\overline{X}] = \sum_{i=1}^{n} \frac{E[X_i]}{n} = \mu$$

and variance

$$\text{Var}(\overline{X}) = \frac{1}{n^2} \sum_{i=1}^{n} \text{Var}(X_i) = \sigma^2/n$$

That is, $\overline{X}$, the average of the sample, is normal with a mean equal to the population mean but with a variance reduced by a factor of $1/n$. It follows from this that

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}$$

is a standard normal random variable.

6.5.2 Joint distribution of $\overline{X}$ and S^2

In this section, we not only obtain the distribution of the sample variance S^2, but we also discover a fundamental fact about normal samples — namely, that $\overline{X}$ and S^2 are independent with $(n-1)S^2/\sigma^2$ having a chi-square distribution with $n-1$ degrees of freedom.

To start, for numbers $x_1, \ldots, x_n$, let $y_i = x_i - \mu$, $i = 1, \ldots, n$. Then as $\overline{y} = \overline{x} - \mu$, it follows from the identity

$$\sum_{i=1}^{n} (y_i - \overline{y})^2 = \sum_{i=1}^{n} y_i^2 - n\overline{y}^2$$

that

$$\sum_{i=1}^{n} (x_i - \overline{x})^2 = \sum_{i=1}^{n} (x_i - \mu)^2 - n(\overline{x} - \mu)^2$$

Now, if $X_1, \ldots, X_n$ is a sample from a normal population having mean μ and variance σ^2, then we obtain from the preceding identity that

$$\frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\sigma^2}=\frac{\sum_{i=1}^{n}(X_i-\overline{X})^2}{\sigma^2}+\frac{n(\overline{X}-\mu)^2}{\sigma^2}$$

or, equivalently,

$$\sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2=\frac{\sum_{i=1}^{n}(X_i-\overline{X})^2}{\sigma^2}+\left[\frac{\sqrt{n}(\overline{X}-\mu)}{\sigma}\right]^2 \tag{6.5.1}$$

Because $(X_i-\mu)/\sigma, i=1,\ldots,n$ are independent standard normals, it follows that the left side of Equation (6.5.1) is a chi-square random variable with n degrees of freedom. Also, as shown in Section 6.5.1, $\sqrt{n}(\overline{X}-\mu)/\sigma$ is a standard normal random variable and so its square is a chi-square random variable with 1 degree of freedom. Thus Equation (6.5.1) equates a chi-square random variable having n degrees of freedom to the sum of two random variables, one of which is chi-square with 1 degree of freedom. But it has been established that the sum of two independent chi-square random variables is also chi-square with a degree of freedom equal to the sum of the two degrees of freedom. Thus, it would seem that there is a reasonable possibility that the two terms on the right side of Equation (6.5.1) are independent, with $\sum_{i=1}^{n}(X_i-\overline{X})^2/\sigma^2$ having a chi-square distribution with $n-1$ degrees of freedom. Since this result can indeed be established, we have the following fundamental result.

Theorem 6.5.1. If $X_1, \ldots, X_n$ is a sample from a normal population having mean μ and variance σ^2, then $\overline{X}$ and S^2 are independent random variables, with $\overline{X}$ being normal with mean μ and variance σ^2/n and $(n-1)S^2/\sigma^2$ being chi-square with $n-1$ degrees of freedom.

Theorem 6.5.1 not only provides the distributions of $\overline{X}$ and S^2 for a normal population but also establishes the important fact that they are independent. In fact, it turns out that this independence of $\overline{X}$ and S^2 is a unique property of the normal distribution. Its importance will become evident in the following chapters.

Example 6.5.a. The time it takes a central processing unit to process a certain type of job is normally distributed with mean 20 seconds and standard deviation 3 seconds. If a sample of 15 such jobs is observed, what is the probability that the sample variance will exceed 12?

Solution. Since the sample is of size $n=15$ and $\sigma^2=9$, write

$$P\{S^2>12\}=P\left\{\frac{14S^2}{9}>\frac{14}{9}\cdot 12\right\}$$

Using R this gives the solution

```
> 1 − pchisq(56/3, 14)
[1] 0.1780811
```
■

The following corollary of Theorem 6.5.1 will be quite useful in the following chapters.

Corollary 6.5.2. Let $X_1, \ldots, X_n$ be a sample from a normal population with mean μ. If $\overline{X}$ denotes the sample mean and S the sample standard deviation, then

$$\sqrt{n}\frac{(\overline{X}-\mu)}{S} \sim t_{n-1}$$

That is, $\sqrt{n}(\overline{X}-\mu)/S$ has a t-distribution with $n-1$ degrees of freedom.

Proof. Recall that a t-random variable with n degrees of freedom is defined as the distribution of

$$\frac{Z}{\sqrt{\chi_n^2/n}}$$

where Z is a standard normal random variable that is independent of χ_n^2, a chi-square random variable with n degrees of freedom. Because Theorem 6.5.1 gives that $\sqrt{n}(\overline{X}-\mu)/\sigma$ is a standard normal that is independent of $(n-1)S^2/\sigma^2$, which is chi-square with $n-1$ degrees of freedom, we can conclude that

$$\frac{\sqrt{n}(\overline{X}-\mu)/\sigma}{\sqrt{S^2/\sigma^2}} = \sqrt{n}\frac{(\overline{X}-\mu)}{S}$$

is a t-random variable with $n-1$ degrees of freedom. ■

6.6 Sampling from a finite population

Consider a population of N elements, and suppose that p is the proportion of the population that has a certain characteristic of interest; that is, Np elements have this characteristic, and $N(1-p)$ do not. A sample of size n from this population is said to be a *random sample* if it is chosen in such a manner that each of the $\binom{N}{n}$ population subsets of size n is equally likely to be the sample. For instance, if the population consists of the three elements a, b, c, then a random sample of size 2 is one that is chosen so that each of the subsets $\{a, b\}$, $\{a, c\}$, and $\{b, c\}$ is equally likely to be the sample. A random subset can be chosen

sequentially by letting its first element be equally likely to be any of the N elements of the population, then letting its second element be equally likely to be any of the remaining $N-1$ elements of the population, and so on.

Suppose now that a random sample of size n has been chosen from a population of size N. For $i = 1, \ldots, n$, let

$$X_i = \begin{cases} 1 & \text{if the } i\text{th member of the sample has the characteristic} \\ 0 & \text{otherwise} \end{cases}$$

Consider now the sum of the X_i; that is, consider

$$X = X_1 + X_2 + \cdots + X_n$$

Because the term X_i contributes 1 to the sum if the ith member of the sample has the characteristic and 0 otherwise, it follows that X is equal to the number of members of the sample that possess the characteristic. In addition, the sample mean

$$\overline{X} = X/n = \sum_{i=1}^{n} X_i/n$$

is equal to the proportion of the members of the sample that possess the characteristic.

Let us now consider the probabilities associated with the statistics X and $\overline{X}$. To begin, note that since each of the N members of the population is equally likely to be the ith member of the sample, it follows that

$$P\{X_i = 1\} = \frac{Np}{N} = p$$

Also,

$$P\{X_i = 0\} = 1 - P\{X_i = 1\} = 1 - p$$

That is, each X_i is equal to either 1 or 0 with respective probabilities p and $1-p$.

It should be noted that the random variables $X_1, X_2, \ldots, X_n$ are not independent. For instance, since the second selection is equally likely to be any of the N members of the population, of which Np have the characteristic, it follows that the probability that the second selection has the characteristic is $Np/N = p$. That is, without any knowledge of the outcome of the first selection,

$$P\{X_2 = 1\} = p$$

However, the conditional probability that $X_2 = 1$, given that the first selection has the characteristic, is

$$P\{X_2 = 1|X_1 = 1\} = \frac{Np - 1}{N - 1}$$

which is seen by noting that if the first selection has the characteristic, then the second selection is equally likely to be any of the remaining $N - 1$ elements, of which $Np - 1$ have the characteristic. Similarly, the probability that the second selection has the characteristic given that the first one does not is

$$P\{X_2 = 1|X_1 = 0\} = \frac{Np}{N - 1}$$

Thus, knowing whether or not the first element of the random sample has the characteristic changes the probability for the next element. However, when the population size N is large in relation to the sample size n, this change will be very slight. For instance, if $N = 1000$, $p = .4$, then

$$P\{X_2 = 1|X_1 = 1\} = \frac{399}{999} = .3994$$

which is very close to the unconditional probability that $X_2 = 1$; namely,

$$P\{X_2 = 1\} = .4$$

Similarly, the probability that the second element of the sample has the characteristic given that the first does not is

$$P\{X_2 = 1|X_1 = 0\} = \frac{400}{999} = .4004$$

which is again very close to .4.

Indeed, it can be shown that when the population size N is large with respect to the sample size n, then $X_1, X_2, \ldots, X_n$ are approximately independent. Now if we think of each X_i as representing the result of a trial that is a success if X_i equals 1 and a failure otherwise, it follows that $X = \sum_{i=1}^{n} X_i$ can be thought of as representing the total number of successes in n trials. Hence, if the X_i were independent, then X would be a binomial random variable with parameters n and p. In other words, when the population size N is large in relation to the sample size n, then the distribution of the number of members of the sample that possess the characteristic is approximately that of a binomial random variable with parameters n and p.

Remark

Of course, X is a hypergeometric random variable (Section 5.4); and so the preceding shows that a hypergeometric can be approximated by a binomial random variable when the number chosen is small in relation to the total number of elements.

For the remainder of this text, we will suppose that the underlying population is large in relation to the sample size and we will take the distribution of X to be binomial.

By using the formulas given in Section 5.4 for the mean and standard deviation of a binomial random variable, we see that

$$E[X] = np \quad \text{and} \quad SD(X) = \sqrt{np(1-p)}$$

Moreover $\overline{X} = X/n$, the proportion of the sample that has the characteristic, has mean and variance given by

$$E[\overline{X}] = E[X]/n = p$$

and

$$\text{Var}(\overline{X}) = \text{Var}(X)/n^2 = p(1-p)/n$$

Example 6.6.a. Suppose that 45 percent of the population favors a certain candidate in an upcoming election. If a random sample of size 200 is chosen, find

(a) the expected value and standard deviation of the number of members of the sample that favor the candidate;
(b) the probability that more than half the members of the sample favor the candidate.

Solution. (a) The expected value and standard deviation of the proportion that favor the candidate are

$$E[X] = 200(.45) = 90, \quad SD(X) = \sqrt{200(.45)(1-.45)} = 7.0356$$

(b) Because X is binomial with parameters 200 and .45, use R to obtain

$$P(X \geq 101) =$$

```
> 1 - pbinom(100, 200, .45)
[1] 0.06807525
```

■

Even when each element of the population has more than two possible values, it still remains true that if the population size is large in relation to the sample size, then the sample data can be regarded as being independent random variables from the population distribution.

Example 6.6.b. According to the U.S. Department of Agriculture's *World Livestock Situation*, the country with the greatest per capita consumption of pork is Denmark. In 2013, the amount of pork consumed by a person residing in Denmark had a mean value of 147 pounds with a standard deviation of 62 pounds. If a random sample of 25 Danes is chosen, approximate the probability that

the average amount of pork consumed by the members of this group in 2013 exceeded 150 pounds.

Solution. If we let X_i be the amount consumed by the ith member of the sample, $i = 1, \ldots, 25$, then the desired probability is

$$P\left\{\frac{X_1 + \cdots + X_{25}}{25} > 150\right\} = P\{\overline{X} > 150\}$$

where $\overline{X}$ is the sample mean of the 25 sample values. Since we can regard the X_i as being independent random variables with mean 147 and standard deviation 62, it follows from the central limit theorem that their sample mean will be approximately normal with mean 147 and standard deviation 62/5. Thus, with Z being a standard normal random variable, we have

$$P\{\overline{X} > 150\} = P\left\{\frac{\overline{X} - 147}{12.4} > \frac{150 - 147}{12.4}\right\}$$

$$P(\bar{X} > 150) \approx P\left(Z > \frac{150 - 147}{12.4}\right)$$

$$> 1 - \text{ pnorm}(3/12.4)$$

$$[1]\ 0.4044151 \quad \blacksquare$$

Problems

1. Suppose that X_1, X_2, X_3 are independent with the common probability mass function

   $$P\{X_i = 0\} = .2, \quad P\{X_i = 1\} = .3, \quad P\{X_i = 3\} = .5, \quad i = 1, 2, 3$$

   a. Plot the probability mass function of $\overline{X}_2 = \dfrac{X_1 + X_2}{2}$.
   b. Determine $E[\overline{X}_2]$ and $\text{Var}(\overline{X}_2)$.
   c. Plot the probability mass function of $\overline{X}_3 = \dfrac{X_1 + X_2 + X_3}{3}$.
   d. Determine $E[\overline{X}_3]$ and $\text{Var}(\overline{X}_3)$.
2. If 10 fair dice are rolled, approximate the probability that the sum of the values obtained (which ranges from 10 to 60) is between 30 and 40 inclusive.
3. Approximate the probability that the sum of 16 independent uniform (0, 1) random variables exceeds 10.
4. A roulette wheel has 38 slots, numbered 0, 00, and 1 through 36. If you bet 1 on a specified number, you either win 35 if the roulette ball lands on that number or lose 1 if it does not. If you continually make such bets, approximate the probability that
   a. you are winning after 34 bets;

b. you are winning after 1000 bets;
c. you are winning after 100,000 bets.

Assume that each roll of the roulette ball is equally likely to land on any of the 38 numbers.

5. A highway department has enough salt to handle a total of 80 inches of snowfall. Suppose the daily amount of snow has a mean of 1.5 inches and a standard deviation of .3 inch.
   a. Approximate the probability that the salt on hand will suffice for the next 50 days.
   b. What assumption did you make in solving part (a)?
   c. Do you think this assumption is justified? Explain briefly.
6. Fifty numbers are rounded off to the nearest integer and then summed. If the individual roundoff errors are uniformly distributed between $-.5$ and .5, what is the approximate probability that the resultant sum differs from the exact sum by more than 3?
7. A six-sided die, in which each side is equally likely to appear, is repeatedly rolled until the total of all rolls exceeds 400. Approximate the probability that this will require more than 140 rolls.
8. The amount of time that a certain type of battery functions is a random variable with mean 5 weeks and standard deviation 1.5 weeks. Upon failure, it is immediately replaced by a new battery. Approximate the probability that 13 or more batteries will be needed in a year.
9. The lifetime of a certain electrical part is a random variable with mean 100 hours and standard deviation 20 hours. If 16 such parts are tested, find the probability that the sample mean is
   a. less than 104;
   b. between 98 and 104 hours.
10. A tobacco company claims that the amount of nicotine in its cigarettes is a random variable with mean 2.2 mg and standard deviation .3 mg. However, the sample mean nicotine content of 100 randomly chosen cigarettes was 3.1 mg. What is the approximate probability that the sample mean would have been as high or higher than 3.1 if the company's claims were true?
11. The lifetime (in hours) of a type of electric bulb has expected value 500 and standard deviation 80. Approximate the probability that the sample mean of n such bulbs is greater than 525 when
   a. $n = 4$;
   b. $n = 16$;
   c. $n = 36$;
   d. $n = 64$.
12. An instructor knows from past experience that student exam scores have mean 77 and standard deviation 15. At present the instructor is teaching two separate classes — one of size 25 and the other of size 64.
   a. Approximate the probability that the average test score in the class of size 25 lies between 72 and 82.

b. Repeat part (a) for a class of size 64.
c. What is the approximate probability that the average test score in the class of size 25 is higher than that of the class of size 64?
d. Suppose the average scores in the two classes are 76 and 83. Which class, the one of size 25 or the one of size 64, do you think was more likely to have averaged 83?

13. If X is binomial with parameters $n = 150$, $p = .6$, compute the exact value of $P\{X \leq 80\}$ and compare with its normal approximation both (a) making use of and (b) not making use of the continuity correction.

14. Teams 1, 2, 3, 4 are all scheduled to play with each of the other teams 10 times. Whenever team i plays team j, team i is the winner with probability $P_{i,j}$, and team j is the winner with probability $P_{j,i} = 1 - P_{i,j}$. If

$$P_{1,2} = .6, \quad P_{1,3} = .7, \quad P_{1,4} = .75,$$
$$P_{2,3} = .6, \quad P_{2,4} = .70, \quad P_{3,4} = .5,$$

a. approximate the probability that team 1 wins at least 20 games.
Suppose we want to approximate the probability that team 2 wins at least as many games as does team 1. To do so, let X be the number of games that team 2 wins against team 1, let Y be the total number of games that team 2 wins against teams 3 and 4, and let Z be the total number of games that team 1 wins against teams 3 and 4.
b. Are X, Y, Z independent?
c. Express the event that team 2 wins at least as many games as does team 1 in terms of the random variables X, Y, Z.
d. Approximate the probability that team 2 wins at least as many games as team 1.

15. A club basketball team will play a 60-game season. Thirty-two of these games are against class A teams and 28 are against class B teams. The outcomes of all the games are independent. The team will win each game against a class A opponent with probability .5, and it will win each game against a class B opponent with probability .7. Let X denote its total number of victories in the season.
a. Is X a binomial random variable?
b. Let X_A and X_B denote, respectively, the number of victories against class A and class B teams. What are the distributions of X_A and X_B?
c. What is the relationship between X_A, X_B, and X?
d. Approximate the probability that the team wins 40 or more games.

16. Argue, based on the central limit theorem, that a Poisson random variable having mean λ will approximately have a normal distribution with mean and variance both equal to λ when λ is large. If X is Poisson with mean 100, compute the exact probability that X is less than or equal to 116 and compare it with its normal approximation both when a continuity correction is utilized and when it is not. The convergence of the Poisson to the normal is indicated in Figure 6.4.

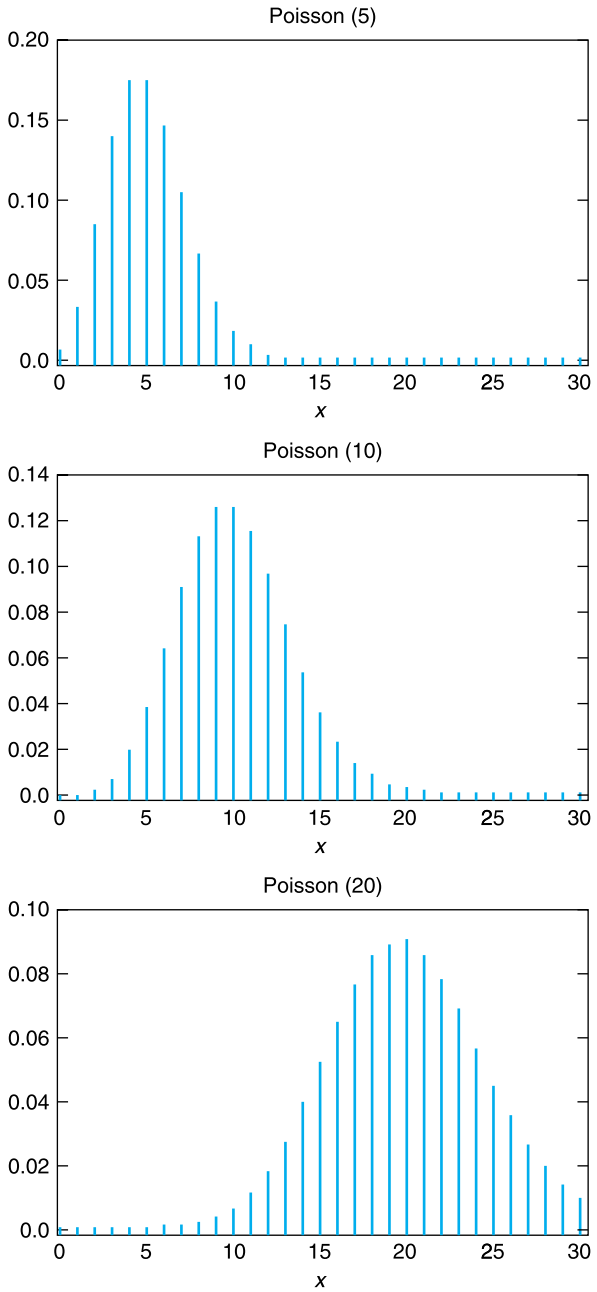


FIGURE 6.4
Poisson probability mass functions.